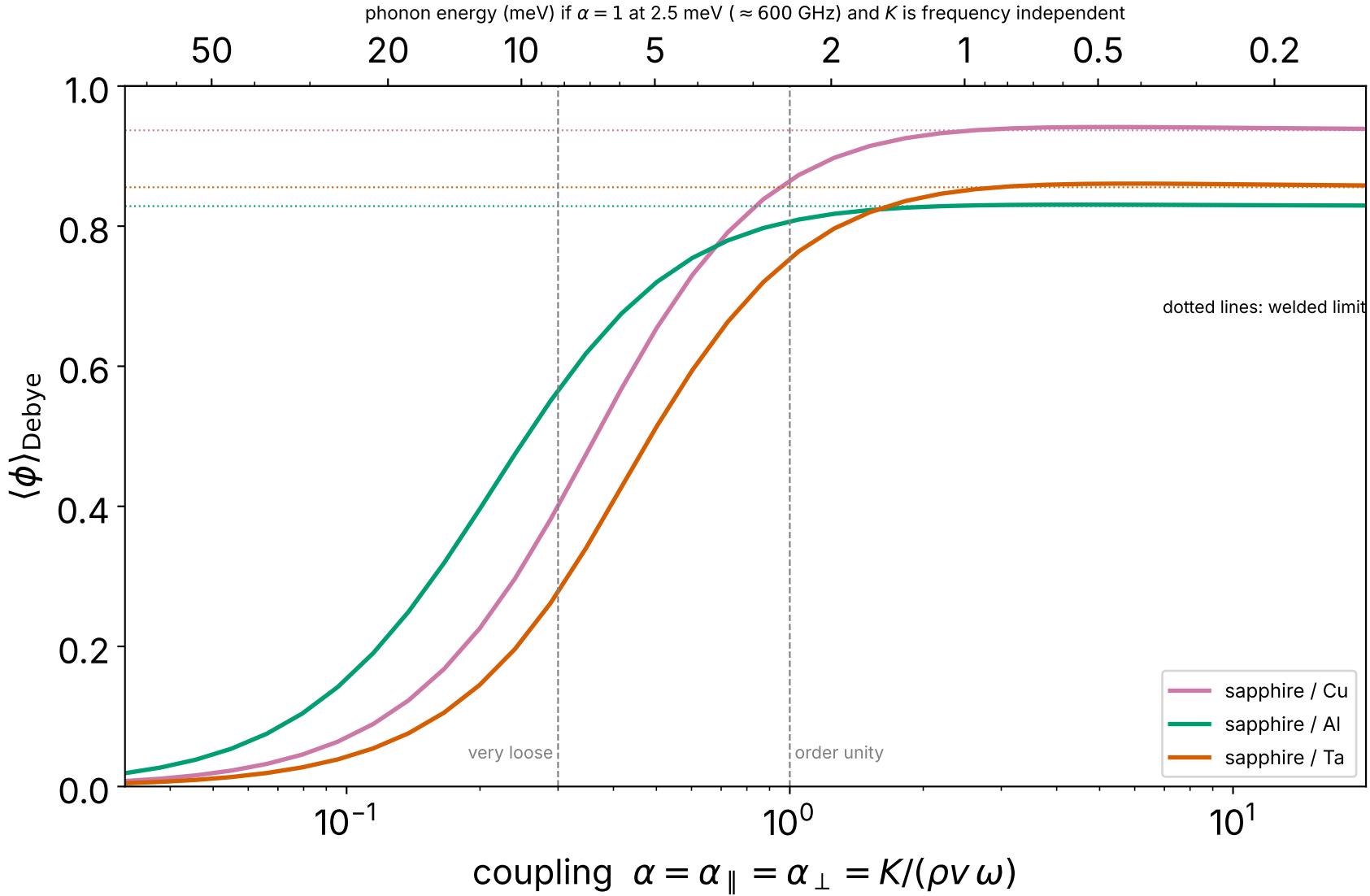
phonon energy (meV) if $\alpha = 1$ at 2.5 meV (≈ 600 GHz) and K is frequency independent
50
20
10
5
2
1
0.5
0.2
1.0
0.8
0.6
0.4
0.2
0.0
$\langle\phi\rangle_{\mathrm{Debye}}$
dotted lines: welded limit
very loose
order unity
sapphire / Cu
sapphire / Al
sapphire / Ta
10^{-1}
10^{0}
10^{1}
coupling $\alpha = \alpha_{\parallel} = \alpha_{\perp} = K/(\rho v\, \omega)$